Is the Universe a Black Hole?

To characterize the scale of the Universe in this model, one must specify either its **radius or its mass**; the other quantity can then be derived from the **Schwarzschild radius**.

I have incorporated this black-hole relationship into a **rotating-Universe model**. The model produces the distance modulus as a function of redshift, as well as the expansion velocity, expansion acceleration, and Hubble parameter.

Interestingly, the calculated Hubble constant depends on whether the model uses the **speed of light, c**, or the calculated **expansion velocity, v**, in the corresponding relation.

UniverseFit27

R0 = 1.3744e+01 light-years

Mass = 8.7547e+52 kg

Theta = 2.0000e-01 rad

Hubble constant (c) = 71.1433 km/s/Mpc

Hubble constant (v) = 76.5536 km/s/Mpc

Expansion velocity = 3.2259e+08 m/s

Expansion acceleration = 1.6788e-07 m/s^2

SN Ia number = 1464

GRB number = 190

SN chi2 = 668.3653

GRB chi2 = 1128.8937

Total chi2 = 1797.2591

Reduced SN chi2 = 0.4565

Reduced GRB chi2 = 5.9415

Reduced total chi2 = 1.0866

The resulting characteristic radius is about **13.74 light-years**, corresponding to a mass of approximately **8.75 × 10^52 kg** when interpreted through the Schwarzschild-radius relation.

The key question is therefore:

Could the observable Universe be described, at least mathematically, as a system whose characteristic radius is comparable to its Schwarzschild radius?

This is an intriguing possibility to investigate, but it is important to distinguish the mathematical similarity from the physical identification of the Universe with a black hole. A Schwarzschild black hole is a solution describing a particular spacetime geometry, whereas the Universe is described by cosmological spacetime models with expansion and, in this case, rotation. The fact that a mass and radius satisfy the Schwarzschild relation does not by itself demonstrate that the Universe is a black hole.

#Cosmology #Physics #Astrophysics #Universe #BlackHole #GeneralRelativity #SchwarzschildRadius #HubbleParameter #HubbleConstant #CosmologyModel #RotatingUniverse #Supernova #GammaRayBursts #DarkEnergy #宇宙 #AstrophysicsResearch